

Vidvathama Ramesh

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Research Interests

Visual navigation and embodied perception; feedforward 3D scene reconstruction and lifelong SLAM; dexterous manipulation; NLP and LLM systems (routing, evaluation, safety); decision theory and strategic reasoning applied to multi-agent and reinforcement-learning systems; energy- and latency-constrained compute for onboard autonomy.

Education

International Institute of Information Technology, Hyderabad (IIIT-H) **July 2024 – Present**
B.Tech + MS by Research (Dual Degree), Electronics and Communication Engineering; CGPA: 9.01/10.0 (9.1+ over the last three semesters); Expected May 2028 *Hyderabad, India*

- **Graduate / Advanced Coursework:** Advanced Natural Language Processing, Digital VLSI Design, Digital Image Processing, Quantum Materials and Devices, Statistical Methods in AI, Introduction to Natural Language Processing, Mobile Robotics, UAV Design, Computer Vision, Introduction to Reinforcement Learning, Robotics Planning and Navigation.
- **Core Coursework:** Processor Architecture, Digital Systems and Logic, Signal Processing, Information Theory, Probability and Random Processes, Data Structures and Algorithms, Functional Analysis, Systems Thinking.

Research Experience

Research Intern — TCS Research **May 2026 – July 2026**
Mentor: Jeripothula Prudviraj *Hyderabad, India*

- Architected deep neural encoders that extract high-level aesthetic and stylistic features as state representations for autonomous reinforcement-learning agents, and shaped policy-gradient reward functions for iterative generative decision-making.
- Engineered a human-AI co-creation framework combining computer vision and RL into real-time agents that adapt to human creative intent; delivered as the **CanvasMind** system (below).
- Outputs: submission to **NeurIPS 2026** (Creative AI Track: Agency) [submitted]; **ACM CHI 2027** manuscript in preparation; research paper scoped for a museum collaboration.

Undergraduate Researcher — Robotics Research Centre (RRC), IIIT-H **May 2025 – Present**
Supervisor: Prof. Sourav Garg *Hyderabad, India*

- Benchmarking feedforward 3D mapping foundation models (DUS_t3R, MAS_t3R, CUT₃R, VGGT, MapAnything, Plann_{3r}) against an in-house visual navigation stack, under a common .npz output schema that normalises pose, point-map and confidence conventions across all five model families.
- Isolated and mitigated the two dominant failure modes at the mapping-navigation seam: metric-scale ambiguity (up-to-scale vs. metric reconstruction, resolved via Umeyama / Sim(3) alignment with evo) and camera frame-convention mismatch (OpenCV vs. robotics conventions), covered by day-one unit tests.
- Converting predicted point-maps into occupancy grids (Open3D) for classical planning and control, and running a geometric-backbone ablation on Plann_{3r} plus a latency-vs-number-of-views scaling study.
- Integrating Vision-Language-Action models (VLAs/VLMs) with topological graph structuring for robust visual navigation in unstructured environments.
- Prototyped 3D-vision + RL policies for dexterous manipulation on a LEAP Hand with tactile simulation via TacSL in NVIDIA Isaac Lab.
- Preparing a first-author submission to **IEEE IROS 2027**.

Research Intern — Indian Institute of Technology Bombay (IIT Bombay) **May 2025 – Aug 2025**
Supervisor: Prof. Vikram Bansal *Mumbai, India*

- Implemented a TransCrowd transformer pipeline for high-density crowd counting, integrating spatio-temporal features to improve density-estimation accuracy across large-scale benchmark datasets.

Publications and Manuscripts in Preparation

- **Feedforward 3D Mapping for Visual Navigation** — first-author manuscript targeting **IEEE IROS 2027**; Robotics Research Centre, IIIT-H (Prof. Sourav Garg).
- **CanvasMind: Multi-Agent Inference-Time RL for Creative AI** — co-authored; **NeurIPS 2026** Creative AI Track (submitted); **ACM CHI 2027** manuscript in preparation.

- **HiGrEC: Hierarchical Classification for Political Question-Evasion Detection** — system paper on the public CLARITY / QEvasion benchmark (SemEval-2026 Task 6), team Nier_ANLP; targeting an ACL-affiliated workshop.
- **Adaptive Semantic Parallelism: A Select-Then-Route Framework for LLM Inference** — Targetting ACL 2027, pipeline traces and an interactive demo.

Selected Projects — Robotics, Perception and Control

Lifelong SLAM for Long-Term Mobile Robot Operation

2026

Mobile Robotics Coursework; Python, C++, ROS, g2o/GTSAM, Open3D

- Built a multi-session SLAM system that maintains a single consistent map across repeated traversals of the same environment, addressing the core lifelong-mapping problem: keeping localisation reliable while the scene itself changes between sessions.
- Implemented pose-graph back-end optimisation with loop-closure detection and robust outlier rejection, enabling relocalisation into and merging of previously built maps rather than rebuilding from scratch each session.
- Added map-maintenance policies - keyframe pruning, dynamic- and semi-static-object handling, and bounded graph growth - so map size and optimisation cost stay tractable over long deployments.
- Reported absolute and relative trajectory error against a single-session baseline and map-size growth with and without pruning.

Quadrotor Trajectory Planning and ROS Simulation

March 2026

Mobile Robotics Coursework; MATLAB, Python, ROS

- Derived acceleration-to-attitude control mappings (thrust, roll and pitch commands) for multi-waypoint quadrotor navigation and validated the resulting trajectories in a ROS simulation.

Gyrocopter (Autogyro) Design, Build and Flight Testing

March 2026

UAV Design (EC4.402); ArduPilot/ArduPlane, XFLR5

- Assembled a free-spinning-rotor autogyro prototype (fixed-wing fuselage, forward-thrust BLDC propulsion) and configured ArduPilot/ArduPlane firmware for stabilisation assistance.
- Performed blade-element and airfoil-selection aerodynamic analysis in XFLR5 and derived flight-dynamics calculations for rotor sizing and stability margins.

Selected Projects — VLSI and Computer Systems

4-bit Carry Look-Ahead Adder — Full-Custom 180 nm CMOS

2026

Digital VLSI Design; NGSPICE, MAGIC (SCN6M_DEEP.09), Icarus Verilog, GTKWave, Saleae Logic 2

- Executed the complete custom-IC flow end to end — logic design, transistor sizing, Euler-path stick diagrams, schematic SPICE, MAGIC layout, parasitic extraction, post-layout timing, structural RTL and FPGA validation — for a synchronous 4-bit CLA adder in static CMOS at 180 nm (TSMC models, $V_{DD} = 1.8\text{ V}$), achieving a **1.25 GHz** maximum reliable clock, 5.393 mW average power, a 3.97 pJ power-delay product, and a $64.08\mu\text{m} \times 86.31\mu\text{m}$ final layout.
- Realised generate ($G_i = A_i \cdot B_i$) and propagate ($P_i = A_i \oplus B_i$) terms as AND/XOR cells and all four carry equations as static-CMOS complex gates so every carry resolves in parallel rather than rippling; registered I/O with TSPC (True Single-Phase Clock) flip-flops to eliminate complementary-clock distribution, characterising them at $t_{setup} = 50\text{ ps}$, $t_{hold} = 30\text{ ps}$, $t_{pcq} = 118\text{ ps}$.
- Closed timing on the parasitic-extracted full-chip netlist — $t_{CLK} \geq t_p + t_{setup} + t_{pcq} = 736\text{ ps}$ on the critical path $a0 \rightarrow s3$, set to 800 ps after observing glitching at that boundary — then cross-validated against structural Verilog with an on-chip clock divider exhausting all 256 input combinations, deployed to a Boolean Board FPGA and captured on a logic analyser.

32-bit RISC-V Processor

May 2025

Processor Architecture; Verilog, Icarus Verilog, Python, RISC-V Assembly

- Designed a 32-bit RISC-V processor in Verilog, scaling from a single-cycle sequential datapath to a 5-stage IF/ID/EX/MEM/WB pipeline: program counter, register file, instruction and data memories, ALU and control, and immediate-generation units for R-, I-, S- and B-type instructions.
- Engineered forwarding and hazard-detection units covering data, double-data, load-store, control and load-use hazards using forwarding paths, stalls and branch flushes.
- Wrote a Python assembler and Icarus Verilog testbenches validating Fibonacci, linear search, vector addition and summation programs.

PolyQuery (PolyRA) — Multi-Model Relational Algebra Engine

Nov. 2025

Data Systems; C++17, Systems Programming, Concurrency

- Engineered a high-performance, integer-optimised C++17 RDBMS unifying relational algebra, graph processing and high-dimensional vector search over a custom block-based buffer manager with external-memory algorithms (external merge sort, external hashing, B-Tree indexing).
- Implemented multi-threaded transactions with Wait-Die concurrency control and page-granular shared/exclusive locking (provably deadlock-free), disk-backed BFS path queries, and KNN search (Euclidean and cosine) via a bounded min-heap — all under a fixed 10-page memory ceiling.

POSIX-Core C Shell

Aug. 2025

Operating Systems; C, Unix Syscalls, Signals, Valgrind

- Developed a modular `execvp`-based Unix shell with foreground/background execution, unlimited pipelines, full I/O redirection and `.myshrc` alias resolution; verified leak-free execution under Valgrind.
- Implemented 12+ built-ins (`hop`, `reveal`, `seek`, `log`, `proclore`, `activities`, `fg/bg`, `neonate`, and an `iMan` TCP man-page fetcher) with robust SIGINT/SIGTSTP/SIGCHLD handling and asynchronous child reaping.

Selected Projects — Natural Language Processing and Machine Learning

HiGrEC — Hierarchical Classification for Political Question-Evasion Detection

2026

Advanced NLP course project; SemEval-2026 Task 6 (CLARITY), Team Nier_ANLP; PyTorch, HuggingFace

- Reframed the task correctly at the outset: evasion is a *pragmatic* property of a question–answer pair (violations of Gricean Relation and Quantity), not deception detection (Gricean Quality) — a distinction that determines what the model must be conditioned on and separates this system from prior lie-detection framings.
- C1 — Taxonomy-constrained hierarchical head:** enforces label consistency *by construction* through deterministic marginalisation over the CLARITY evasion taxonomy, removing the parent–child contradictions that flat multi-class baselines produce.
- C2 — Question-demand auxiliary supervision:** supervises the model on what the question actually *demand*s, with long-input chunking for extended parliamentary exchanges; ModernBERT used as a controlled baseline to disentangle supervision gains from raw context-length gains.
- C3 — Disagreement-aware training:** trains on raw annotator distributions as soft labels plus a confusion-aware contrastive objective, attacking representational overlap between adjacent evasion types rather than treating the problem as class imbalance.
- C4 — Post-hoc macro-F1-aware calibration:** low-cost decision-threshold calibration tuned directly for the macro-F1 metric the benchmark scores on.
- Ran the project against explicit **go/no-go gates**: every component must clear a threshold in out-of-fold macro-F1 across three seeds or be cut, with failures written up as negative results. Reference systems targeted: TeleAI (macro-F1 0.68) and CLaC (0.59).

Adaptive Semantic Parallelism (ASP) — Select-Then-Route LLM Inference

Jan. 2026

Introduction to NLP course project; DistilBERT, PyTorch, LightGBM, spaCy

- Architected an 8-stage LLM routing pipeline that decomposes a prompt into a semantic DAG (spaCy, SentenceTransformer, constituency parsing) and routes each sub-task to a strong or weak model via a 27-feature LightGBM router with BFS strong-inheritance — **85.66% routing accuracy, 88.04% F1, ~49% inference-cost reduction at 131 ms mean latency**.
- Fine-tuned a dual-head DistilBERT (11-class intent classification + complexity regression; 83.32% accuracy, 85.32% F1) as the front-end selector.
- Built an asynchronous DAG execution engine with speculative execution, semantic caching and a shared context blackboard, yielding 60–80% token savings.
- Layered a 3-stage safety, NLI-verification and LLM-as-a-Judge stack over routing and aggregation; shipped evaluation scripts, pipeline traces and an interactive demo alongside an ACL-format paper.

CanvasMind — Multi-Agent Inference-Time RL for Creative AI

May 2026

TCS Research; FastAPI, Azure OpenAI, Python

- Built a multi-agent creative-AI system in which generative agents with distinct personas co-paint a shared canvas under a critic agent, wrapped in an inference-time RL layer using best-of-N search, a batched reward model, a persistent UCB1 bandit and reward-triggered Reflexion self-correction.
- Modelled agent interaction with misaligned per-agent rewards as a general-sum game; applied Shapley-value attribution, empowerment and Goodhart diagnostics to identify and correct four distinct RL failure modes, lifting composite quality from -0.05 to $+1.66$ at equal compute across a 4,000-session ablation.

Sequence Modelling — Cipher Decryption and Masked Language Modelling

Jan. 2026

Introduction to NLP; PyTorch, HuggingFace, Weights & Biases

- Built RNN and LSTM models from scratch (no `nn.RNN/nn.LSTM`) for character-level cipher decryption, and SSM/BiLSTM models for masked language modelling; tracked all experiments in Weights & Biases and published trained models to HuggingFace (Vidvathama).

Transformers, CNNs and Optimisation — Architecture and Ablation Study

Dec. 2025

Statistical Methods in AI; PyTorch

- Implemented a decoder-only Transformer comparing Pre-Norm and Post-Norm configurations, a CNN with Grad-CAM interpretability, and a controlled optimizer/normalisation ablation study on CIFAR-10.

Technical Skills

Programming Languages: C, C++ (C++17), Python, Verilog, RISC-V Assembly, x86 Assembly, JavaScript, SQL, Bash, MATLAB, $\text{\LaTeX}$

Machine Learning: PyTorch, TensorFlow, HuggingFace Transformers, Scikit-Learn, LightGBM, spaCy, NumPy, Pandas, Weights & Biases, Grad-CAM

Robotics and Vision: ROS 2, NVIDIA Isaac Sim / Isaac Lab, TacSL, Open3D, OpenCV, evo, ArduPilot / ArduPlane, XFLR5, SLAM and pose-graph optimisation (g2o / GTSAM), point-cloud and occupancy-grid processing

VLSI and Hardware: NGSPICE, MAGIC VLSI (SCN6M_DEEP.09), Icarus Verilog, GTKWave, Saleae Logic 2, FPGA prototyping (Boolean Board), full-custom layout, parasitic extraction, static timing and power analysis

Systems: Operating systems, Linux/Unix, POSIX, system calls, signals, concurrency and multithreading, parallel computing, CPU architecture, digital systems design

Web, Backend and Data: FastAPI, Flask, LangGraph, Azure OpenAI, Next.js, React, Node.js, PostgreSQL, MySQL

Tools: Git, Docker, Valgrind, GDB

Achievements and Honours

- **Dean's List for Academic Excellence**, IIIT Hyderabad — top 5% of batch (Spring, UG2) and top 10% of batch (Monsoon, UG3).
- **1st Prize, SUD Life Bank Hackathon** — built an intelligent insurance platform (FinTech / banking track).
- **Top 10 Finalist, Lloyds IP&I Hackathon (2025)** — voice-first IVR replacement using retrieval-augmented generation (Whisper, GPT-4o, FAISS) to ground responses in policy documents and auto-file tickets; separately recognised at the Lloyds Banking Agentic AI Hackathon.